Report 1: Project Description

Library Management System (LMS)

Course / Institution	Kigali Independent University (ULK)
Assignment	UML Class Diagram Modelling Assignment
Instructor	Eric Maniraguha
Group Member 1	[Full Name] - Student ID: [_____]
Group Member 2	[Full Name] - Student ID: [_____]
Submission date	20 October 2026

1. Project Title and Business Idea

Project title: Library Management System (LMS).
Business idea: A school or community library needs an information system to register members, catalogue books, record book loans and reservations, and calculate fines for late returns. The system replaces paper registers and reduces lost books, long queues and calculation mistakes.

2. Problem Statement and Purpose

Many libraries still record loans in notebooks. This causes several problems:

• It is difficult to know which books are available or who currently holds them.

• Late returns are not tracked, so fines are calculated manually and inconsistently.

• Members cannot reserve a book that is already borrowed.

• Searching for a book by title, author or category takes a long time.

• Reports for the library management are slow and error-prone.

Purpose: to design a clear UML class model of a library system that organises books, members, loans, reservations and fines, so that a future application can be built from it.

3. Project Scope

Functions covered (in scope)

• Member registration and member information management.

• Book catalogue management: books, authors and categories.

• Borrowing and returning books (loans) with due dates.

• Book reservations when a book is unavailable.

• Fine calculation and payment for overdue loans.

• Librarian operations: registering members, adding books, issuing loans and processing returns.

Functions excluded (out of scope)

• Online payment gateways and mobile-money integration.

• E-books and digital downloads.

• Inter-library loans and purchasing or supplier management.

• Building a fully functional application (this project only designs the UML class diagram).

• Staff payroll and accounting.

4. Main Classes and Their Responsibilities

Class	Responsibility
Person (abstract)	Holds common data (id, name, email, phone) shared by all users of the system.
Member	A registered library user who borrows books, returns them and makes reservations.
Librarian	Staff member who registers members, adds books, issues loans and processes returns.
Book	A title in the catalogue with its stock information and availability.
Author	Writer of one or more books.
Category	Groups books by subject (e.g. Science, Fiction, History).
Loan	Records that a member borrowed a book, with loan date, due date and return date.
Reservation	Records a member's request to hold a book that is not currently available.
Fine	Penalty generated when a loan is returned late.

5. System Requirements

Functional requirements

• FR1: The system shall register, update and view member records.

• FR2: The system shall add and update books, authors and categories.

• FR3: The system shall allow a librarian to issue a book to a member and set a due date.

• FR4: The system shall record the return of a book and update the available copies.

• FR5: The system shall allow a member to reserve an unavailable book.

• FR6: The system shall calculate a fine when a book is returned after its due date.

• FR7: The system shall record fine payments.

Non-functional requirements

• Usability: simple screens that a librarian can learn quickly.

• Security: only librarians can issue loans and edit the catalogue.

• Reliability: loan and stock data must stay consistent.

• Performance: searching a book should take only a few seconds.

6. Assumptions and Business Rules

• A member can have many loans and many reservations over time.

• Each loan is for one book; borrowing several books creates several loans.

• A book can have one or more authors and belongs to one category.

• A fine exists only because of a late loan (it cannot exist without a Loan).

• A loan period is 14 days by default.